

Latest advances in VCSEL technology for next-generation data center network [Invited]

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Directly modulated vertical-cavity surface-emitting lasers (VCSELs) have long dominated the data center optical interconnect market, attributed to VCSELs' cost-effectiveness and energy efficiency. However, numerous challenges are currently being addressed to enhance device characteristics and modulation speed, aligning with the evolving trends in next-generation data centers that underscore the need for enhanced communication rates and extended transmission distances. This review delves into VCSEL structure design, methods for achieving single-mode control, techniques for fabricating long-wavelength VCSELs, and advanced packaging technology for VCSEL arrays, offering insights into recent research on high-speed single-mode VCSEL development and related technologies.

Keywords: directly modulated VCSELs; transverse mode control; optical packaging.

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1. Introduction

Directly modulated vertical-cavity surface-emitting lasers (VCSELs) have held a dominant position in the data center short-reach interconnect market for decades^[1–5], and this supremacy is expected to persist^[6,7]. The datacom VCSELs market is anticipated to surpass 2 billion USD by the year 2027^[8], which is largely attributed to the increasing demand from newly established data centers. The enduring preeminence of VCSEL technology in data center networks (DCNs) can be attributed to its numerous advantages, primarily cost-effectiveness, and energy efficiency^[9,10].

Amid VCSELs' remarkable dominance in data center interconnects, it is important to acknowledge the new advancements that are taking place in DCNs. In recent years, the relentless growth in data traffic, fueled by diverse cloud applications such as video-on-demand, gaming, cloud computing, and artificial intelligence, has driven the upgrading of existing network infrastructure and the new deployment within DCNs. Traditional private and corporate data centers have transitioned into hyperscale data center clusters^[11,12]. On the one hand, the data rate for interconnections within DCNs is steadily progressing towards 400 Gbps^[13–15], while the reach of intradata center (intra-DC) links is expanding, typically spanning distances of up to 2 km^[16]. On the other hand, inter-DC links, facilitating seamless

communication and data synchronization among dispersed data centers, are expected to further increase in significance^[17,18]. These links often cover distances of approximately 10 km within a data center cluster.

To meet the evolving demands of next-generation DCNs, hyperscale data center clusters call for advanced high-speed, high-capacity, and long-distance optical interconnects. Intensity modulation and direct detection (IM/DD) is often considered the primary choice for intra- and inter-DC applications due to its notable advantages in power efficiency, compact footprint, and cost-effectiveness when compared to digital coherence methods^[19,20]. Four-level pulse amplitude modulation (PAM-4), combined with external modulated lasers (EMLs) and coarse wavelength division multiplexing (CWDM) in the O-band, was standardized for contemporary 400 Gbps optical links, capable of covering distances of up to 10 km in DCNs^[21,22]. However, the technical roadmap for achieving data rates of 800 G, 1.6 T, and beyond in DCNs remains uncertain. The escalating data rates exacerbate receiver power sensitivity due to the widening of the receiver noise bandwidth^[23]. Furthermore, optical channel impairments introduce additional challenges in managing the link budget^[24,25]. In this review, the choice between IM/DD and digital coherence^[26], especially for distances of up to 10 km, becomes a vital factor to weigh when addressing the power consumption concerns of DCNs^[27].

A promising solution to tackle the ever-present issue of energy efficiency for 800 G and 1.6 T IM/DD technology is to replace the EML with directly modulated VCSEL, the energy efficiency of which can be as low as 100 fJ/bit^[28,29]. Considerable research efforts have been dedicated to achieving 100 Gbps/lane data transmission by employing PAM4 modulation and 850 nm VCSEL for 800 G Ethernet, targeting a 100-m transmission distance^[30,31]. Furthermore, by reducing an oxide aperture and integrating a mode filter, advances have been made in 850 nm single-mode VCSEL technology to achieve an extended transmission range of around 1 km^[32,33]. Nevertheless, a significant challenge persists in the quest for achieving single-mode operation and longer wavelength capabilities in VCSEL technology for optical links above 1 km. Overcoming this challenge involves the optimization of VCSEL design and performance, marking a crucial frontier in advancing next-generation high-speed optical interconnect technology.

This paper provides a comprehensive review of recent research on high-speed single-mode VCSEL development and related technologies. Section 2 delves into the advancements in VCSEL structure design to improve performance. In Sec. 3, we focus on methods for achieving single-mode control of VCSEL. Section 4 outlines various techniques for fabricating long-wavelength VCSEL. In Sec. 5, we offer an overview of advanced packaging technology for VCSEL arrays aimed at developing large-capacity optical interconnect technology.

2. The Evolution of VCSEL Structure and Design

VCSELs have undergone significant structural evolution over the years since their initial proposal in 1977 by Professor Iga from the Tokyo Institute of Technology^[34]. A modern VCSEL features an architecture that comprises a substrate, distributed Bragg reflectors (DBRs), an active region, and an oxide layer, as illustrated in Fig. 1. The active region typically consists of multiple layers of quantum wells (QWs) and is sandwiched between the top and bottom DBRs, which act as highly reflective mirrors. The selective oxide layer was introduced in the late 1990s to replace the earlier technique of proton implantation for achieving current injection confinement in VCSELs^[35,36]. The original drive behind VCSEL's invention of a fully

monolithic laser cavity fabrication has led to the great commercial success of 850 nm GaAs-based VCSELs incorporating AlGaAs DBRs, AlGaAs/GaAs QWs, and an AlAs oxide layer, with one-step epitaxial growth achieved on GaAs substrates through MOCVD for mass production^[37].

Increasing the modulation speed at a low power consumption level became a critical issue as VCSELs gained popularity in the datacom market. The performance of VCSELs in data transmission is significantly influenced by their dynamic and modulation characteristics. These critical features are comprehensively analyzed through rate equations, which map out the fluctuations and interactions of electrons and photons within the laser cavity over time^[38,39]. The modulation transfer function, also referred to as the intrinsic modulation response, relates the photon density fluctuations to the modulating current obtained as

$$H_{in}(\nu) = \frac{A\nu_r^2}{4\pi^2(\nu_r^2 - \nu^2) + i2\pi\gamma\nu}, \quad (1)$$

where A is a fitting constant, ν_r is the relaxation resonance frequency, and γ is the damping factor. It is evident from the equation that increasing ν_r and reducing γ can effectively broaden the modulation bandwidth of VCSELs. This principle has led some researchers to introduce the terms D factor and K factor^[40,41,42], given that the relaxation resonance frequency and the damping factor can be represented as

$$\nu_r = D\sqrt{(I - I_{th})}, \quad (2)$$

$$\gamma = K\nu_r^2 + \gamma_0, \quad (3)$$

where I_{th} is the threshold current and γ_0 is the damping factor offset. The D factor and K factor are determined by the material and structural parameters of VCSELs, and their expressions are

$$D = \frac{1}{2\pi} \sqrt{\frac{n_i \Gamma v_g \bar{g}}{q V_a \chi}}, \quad (4)$$

$$K = 4\pi^2 \left(\tau_p + \frac{\epsilon \chi}{v_g \bar{g}} \right), \quad (5)$$

where n_i is the internal quantum efficiency, Γ is the confinement factor, v_g is the group velocity, V_a is the volume of the active region, $\bar{g}$ is the differential gain, τ_p is the photon lifetime, ϵ is the gain suppression coefficient, and χ is the transport factor.

Based on Eqs. (4) and (5), it can be deduced that enhanced optical confinement, greater differential gain, and reduced aperture size contribute to an increase in the D factor, while a shorter photon lifetime leads to a decrease in the K factor, resulting in expanded intrinsic modulation bandwidth. It should be noted that thermal effects also influence the intrinsic modulation bandwidth due to the temperature dependence of differential gain and photon saturation within the VCSEL's cavity^[43].

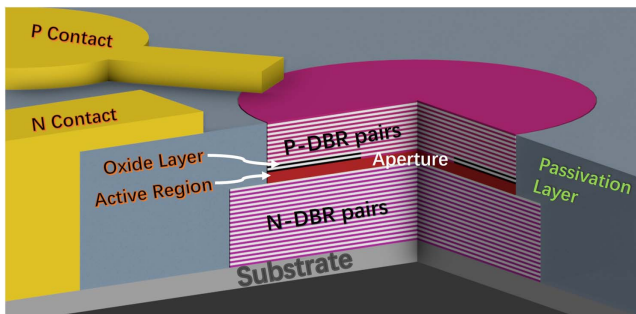


Fig. 1. Cross view of the VCSEL structure.

Despite Eq. (4), the relaxation resonance frequency can also be described in terms of the photon density inside the VCSEL's cavity, as shown in Eq. (6),

$$\nu_r = \frac{1}{2\pi} \sqrt{\frac{S_{\text{avg}} \nu_g \Gamma \tilde{g}}{\tau_p}}, \quad (6)$$

where S_{avg} represents the average photon density. Since the output power P is proportional to S_{avg} , the relaxation resonance frequency is ideally proportional to $\sqrt{P}$. However, from Eq. (3) we observe that the damping factor increases quadratically with ν_r , or linearly with P . Thus, in practice, a trade-off typically exists between the VCSEL's output power and its modulation bandwidth. Given that the resonant frequency, and consequently the bandwidth, increases proportionally to the root of photon density in a low damping scenario, distributing photons across multiple mutually incoherent modes hinders bandwidth enhancement. To approach the intrinsic performance limits of VCSELs, the laser should be designed to operate in a single mode with the highest photon density.

In addition, the overall modulation response of VCSELs should include the extrinsic modulation response, characterized by the parasitic modulation transfer function $H_p(\nu)$. This function can be simplified to resemble a single-pole low-pass filter transfer function^[44], as follows:

$$H_p(\nu) = \frac{B}{1 + i(\nu/\nu_p)}. \quad (7)$$

In total, the small-signal modulation response of VCSELs is approximated by the fit function $H(\nu)$ as

$$H(\nu) = H_{\text{in}}(\nu)H_p(\nu) = \frac{A\nu_r^2}{4\pi^2(\nu_r^2 - \nu^2) + i2\pi\gamma\nu} \times \frac{B}{1 + i(\nu/\nu_p)}, \quad (8)$$

where B is a fitting constant and ν_p is the parasitic cutoff frequency. From Eq. (8), it is clear that besides the intrinsic modulation bandwidth, VCSELs also encounter an RC-limitation stemming from parasitic resistance and capacitance. Parasitic resistance mainly comes from the DBRs, active region, and contacts. On the other hand, the primary sources of parasitic capacitance include the pad capacitance between the metal pad and the bottom mirror stack, as well as capacitance due to oxide layers. The presence of these parasitic elements necessitates careful design and optimization to mitigate their effects, enabling the VCSEL to achieve its potential high-speed performance.

Taking into account both the intrinsic and extrinsic limitations affecting the modulation bandwidth of VCSELs, a series of optimizations have been implemented throughout the development history of VCSELs to enable high-speed operation. P-type modulation doping was applied in the VCSEL design to reduce electrical resistance while maintaining low cavity losses^[45,46]. This approach allows for more efficient current flow, reduced energy losses, and decreased heat generation,

resulting in highly efficient laser operation and improved high-speed modulation capabilities^[47]. The incorporation of strained InGaAs QWs in VCSELs is another effective method for performance optimization. The strain-induced effect splits the light and heavy valence bands while reducing the effective hole mass^[48], which leads to a reduction in the density of states and an increase in the differential gain. Consequently, this method lowers the threshold current and enhances the modulation bandwidth of VCSELs^[49–51]. For the purpose of increasing the differential mode gain, a half-lambda cavity design in VCSELs was adopted to reduce the mode volume^[52,53]. This cavity design features anti-waveguiding behavior, ensuring effective optical confinement of the vertical mode and simultaneous suppression of the in-plane optical field in the active region^[54]. To reduce the photon lifetime, shallow-surface etching is performed on the top of VCSELs^[55,56]. The etching is usually carried out by an inductively coupled plasma (ICP) etcher to achieve a low, stable etch rate. As the etch depth increases, the reflections at the semiconductor/air interface grow increasingly out-of-phase. A shallow etching decreases the top mirror's reflectivity, consequently reducing the photon lifetime within the device. According to Eq. (5), this shorter photon lifetime contributes to a lower K factor. The technique facilitates the extension of the modulation bandwidth by lowering the damping, which results from a reduction in the K factor. However, it should be noted that excessively low damping can lead to jitter issues during large signal modulation^[57]. To tackle the issue of parasitic capacitance, the introduction of a multi-oxide layer VCSEL structure was proposed, effectively aiming to lower the parasitic capacitance^[58,59]. These multiple oxide layers increase the equivalent capacitor thickness, leading to reduced capacitance and increased modulation bandwidth while leaving the fabrication process unchanged^[60]. Surpassing the relaxation resonance frequency limitations in VCSELs is made possible by integrating optical feedback cavity laterally with the VCSEL lasing cavity, which introduces the photon-photon resonance (PPR) effect^[61,62]. The PPR effect can extend the carrier-photon resonance (CPR) frequencies beyond traditional boundaries, which effectively broadens the modulation bandwidth. Besides structural and fabrication optimizations, enhancing the bit rate of VCSELs with limited frequency bandwidth can be achieved through advanced modulation techniques like pulse amplitude modulation (PAM)^[63] and discrete multitone (DMT)^[64,65] to increase spectral efficiency. Additionally, digital equalization techniques are crucial for compensating the VCSELs' transient nonlinearity and bandwidth limitations, adjusting the signal to correct distortions and improving signal quality^[66,67]. Together, these strategies form a comprehensive approach to advancing the performance of VCSELs, facilitating higher data transmission rates that are crucial for DCN.

While the traditional full-cavity VCSEL, with both top and bottom AlGaAs DBRs, is known for its efficient light confinement and emission, the "hybrid-cavity VCSEL" represents an alternative design. In this design, as illustrated in Fig. 2, the top DBRs are replaced with high-contrast grating (HCG) mirrors^[68] or dielectric DBRs^[69], offering a tailored solution for

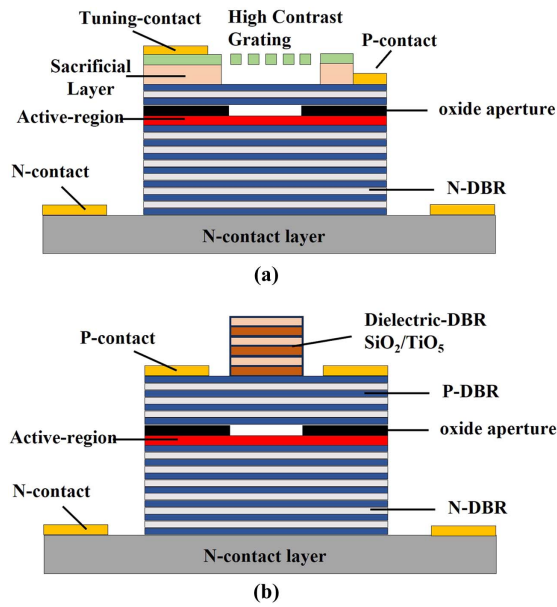


Fig. 2. Schematic view of (a) an HCG VCSEL and (b) a VCSEL with dielectric top DBRs.

specific technological demands like wavelength tuning or silicon integration. The hybrid-cavity VCSEL design with dielectric DBRs is also crucial for long-wavelength VCSEL fabrication, as the QWs suitable for long wavelengths are incompatible with full cavity growth on an InP or GaAs substrate^[70,71]. Dielectric DBRs, constructed from film stacks such as a-Si/SiNx, SiO₂/SiNx, or SiO₂/TiO₅, offer greater index contrast than semiconductor DBRs, allowing for efficient VCSEL lasing, with only a few pairs required^[72,73]. Additionally, a key advantage of this design lies in the post-epitaxial growth deposition of dielectric DBRs, which significantly enhances the adaptability in device design. This allows for the insertion of transverse mode control structures before applying dielectric DBRs^[74], demonstrating the hybrid-cavity VCSEL's adaptability and effectiveness in optimizing laser performance.

3. Transverse Mode Controls in VCSELs

Chromatic dispersion restricts the transmission range of high-speed optical signals in fibers^[75,76], necessitating the use of single-mode light sources to mitigate its impact in long-distance applications. While VCSELs typically can achieve single-longitudinal-mode operation due to their short optical cavities^[77], their ability to maintain stable and efficient single-fundamental-mode (SFM) operation is challenging. The difficulty in achieving SFM operation is particularly evident in VCSELs with larger apertures, required for higher power outputs, due to their susceptibility to excite higher-order transverse modes^[78,79]. Given the importance of single-mode VCSELs in improving transmission quality for long-haul optical communications, such as inter-DC optical links, many research efforts are dedicated to suppressing higher-order transverse modes in

VCSELs, primarily through strategies aimed at elevating the modal losses of the higher-order modes. Since the field distributions inside the cavity differ for various modes, leading to shifts in the intensity profiles of higher-order transverse modes from the central active region, as seen in Fig. 3, introducing a lossy peripheral region in the VCSEL can create a loss gap between the fundamental mode and higher-order modes.

Impurity-induced disordering of AlGaAs DBRs serves as a practical approach for tailoring loss in targeted regions, effectively modifying the energy gap and refractive index in the heterostructures^[80,81]. In Ref. [82], shallow Zinc diffusion on the top DBRs' peripheral region achieves impurity-induced disordering, reducing lateral reflectivity and thereby increasing mirror losses for higher-order transverse modes, while also enhancing optical free-carrier absorption due to an increased free-hole concentration, as shown in Fig. 4(a). This method enabled achieving a single-mode power of 6.5 mW with an SMSR of over 30 dB at 850 nm. Following this approach, the same research group reported a single-mode power of 7.1 mW at 940 nm in VCSELs^[83]. Surface relief in VCSELs, as shown in Fig. 4(b), can also effectively alter mirror loss due to the contact layer's closeness to the active region, whereas even shallow etching can significantly change top reflectivity. In Ref. [84], this approach enabled a multijunction VCSEL to achieve a single-mode output power of 12.55 mW at 904 nm.

Enhancing the absorption loss of higher-order modes is another effective method for enabling SFM lasing in VCSELs, which can be simply accomplished by extending the P electrode partially over the oxide aperture on the top surface^[85,86]. The metal aperture intercepts the beam from the oxide aperture, enhancing absorption at the metal surface, and phase-mismatched reflectivity at the contact/semiconductor interface increases mirror losses for higher-order modes, as shown in Fig. 5. In Ref. [87], a ring-shaped self-aligned recessed metal (SARM) mode filter for transverse mode control was demonstrated. This structure, achieved through selective etching of the top mirror and deposition of p-type contact metals, acts as a spatial filter to suppress higher-order transverse modes. A single-mode power of 1.55 mW and an SMSR of 30.7 dB at 850 nm is achieved.

The use of frequency-selective optical feedback, as explored in Refs. [88,89], has been established as an effective method for

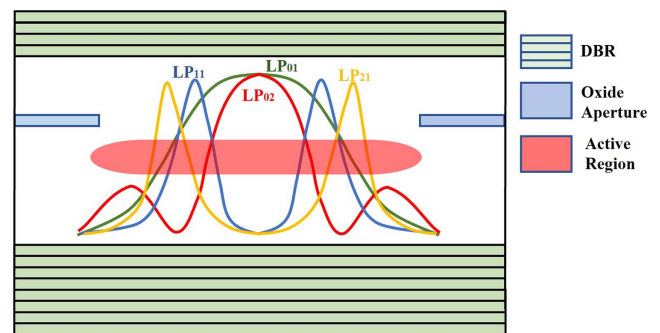


Fig. 3. Field distributions of different transverse modes inside VCSEL cavity.

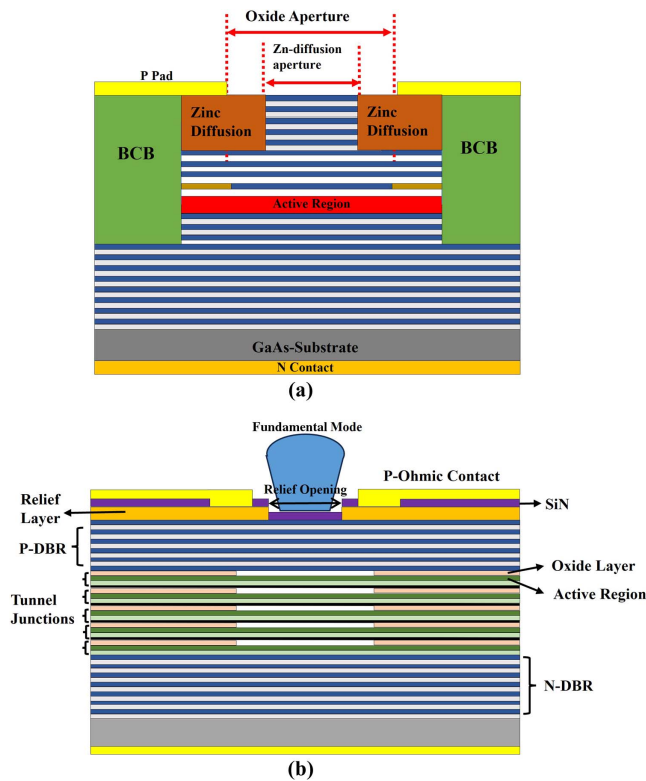


Fig. 4. Schematic of the VCSELs with (a) zinc-diffusion on top mirrors and (b) the multijunction VCSEL with surface relief. BCB, benzocyclobutene.

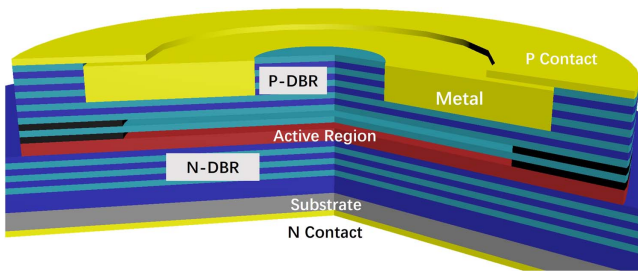


Fig. 5. Schematic of the single-mode VCSEL utilizing P-contact metal as a spatial mode filter.

mode selection in external cavity lasers. This is accomplished by optically coupling one face of the laser to a mode-selective external cavity. A similar effect has been observed in VCSELs with external cavities, as noted in Ref. [90]. However, the challenge in this approach lies in the compact integration of an external cavity with the VCSEL structure. The research team at the Tokyo Institute of Technology has successfully created a hybrid-cavity VCSEL with transverse coupled cavities, where the space between the metal electrode and oxide aperture forms an ultrashort external cavity, which was named a “metal aperture” by the author, as illustrated in Fig. 6. This design has resulted in achieving a single-mode power of 0.5 mW and an SMSR of over 30 dB at 1050 nm^[91].

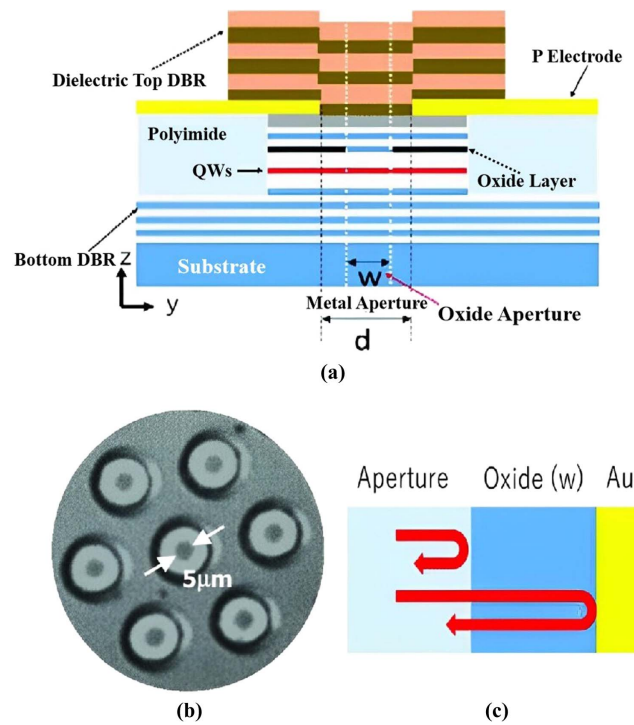


Fig. 6. (a) Schematic of the metal-aperture (MA) VCSEL; (b) photo of the fabricated devices; (c) principle of the optical feedback. Reprinted with permission from Ref. [91]. 2021, IEEE.

4. Long-wavelength VCSELs

In addition to single-mode operation, the emitting wavelength selection of VCSELs is an important factor in reducing both chromatic dispersion and fiber loss. Notably, at the 850 nm wavelength, fiber loss exceeds 1.5 dB/km within a standard G.652 single-mode fiber (SMF), making long-distance transmission of optical signals impractical. To address this challenge, the utilization of long-wavelength single-mode VCSELs is necessary, as longer-wavelength VCSELs can effectively reduce fiber loss and chromatic dispersion, making them ideal for ensuring efficient and reliable signal transmission over kilometer-scale distances^[92–95].

The development of long-wavelength VCSELs at 1.3–1.55 μm faces great challenges due to InP-based materials posing compatibility issues with VCSEL DBR structures. For instance, achieving a 99.9% reflectivity DBR mirror with a 54 nm stopband width at 1.3 μm requires 63 pairs of AlGaInAs/InP grown on an InP substrate due to their small refractive index contrast. In contrast, using just 23 AlGaAs/GaAs pairs can achieve the same reflectivity with a much larger stopband width of 158 nm^[96]. In recent years, significant progress has been made by research institutions and technology companies in addressing these challenges. InP-based VCSELs emitting at 1.5 μm with two dielectric DBRs were reported in Ref. [97]. The active region of this VCSEL contains AlGaInAs/InP QWs and is positioned between an n-doped InP layer and a highly p-doped AlInAs cladding. The semiconductor DBR is replaced by a 5-pair dielectric top AlF_3/ZnS DBR and a 3.5-pair hybrid AlF_3/ZnS -Au

bottom DBR, both with high reflectivities. The proposed VCSEL utilizes a p + AlGaInAs/n + GaInAs buried tunnel junction (BTJ) for current confinement, replacing the AlAs oxide layer in a conventional VCSEL, thereby achieving high conductivity within a circular area, as shown in Fig. 7. 50 Gbps data transmission using PAM4 format was demonstrated over a standard 100 m SMF link.

A technique called wafer fusion was proposed to fuse InP active regions with AlGaAs/GaAs DBRs, enabling the fabrication of 1300 nm VCSELs^[98] and 1550 nm VCSELs^[99]. Initially, this approach faced challenges related to optical losses and device heating due to poor fusion quality. However, ongoing advancements in fabrication techniques and VCSEL structure optimization have led to significant improvements^[100–102]. Russian researchers from the Ioffe Institute and ITMO University reported a 20-Gbps 1300-nm range wafer-fused VCSEL^[103]. The device, depicted in Fig. 8, featuring a 5 μm BTJ diameter, demonstrates stable single-mode lasing and, using a five-tap feedforward equalizer, enables non-return to zero (NRZ) transmission at 25 Gbps over a 5 km distance.

GaAs-based long-wavelength VCSELs present a promising alternative to InP-based ones, due to their compatibility with lattice-matched AlGaAs/GaAs pairs and native oxide layers^[104]. For long-wavelength VCSELs monolithically grown on GaAs substrates, the active regions typically comprise dilute nitride QWs and strained InGaAs. The primary difference in creating a dilute nitride VCSEL compared to conventional types involves the addition of a small amount of nitrogen (N) to the GaAs or

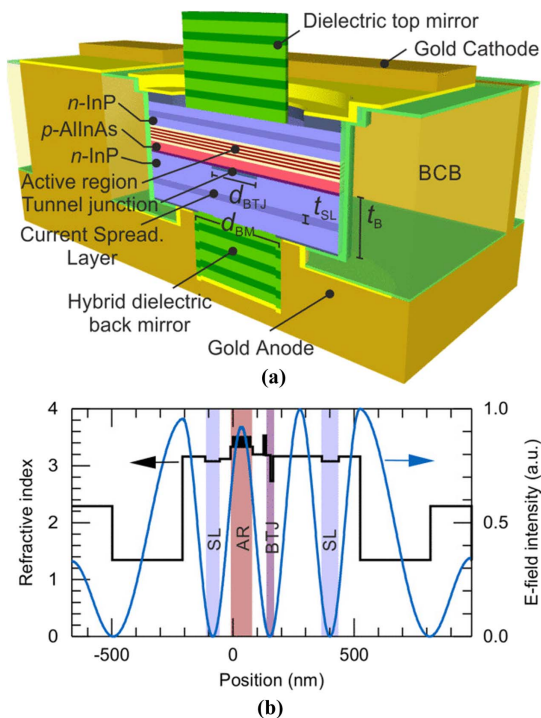


Fig. 7. (a) Schematic of an InP-based 1.5 μm VCSEL with BTJ and dielectric DBRs; (b) refractive index profile and field intensity distribution of the device. Reprinted with permission from Ref. [97]. 2017, IEEE.

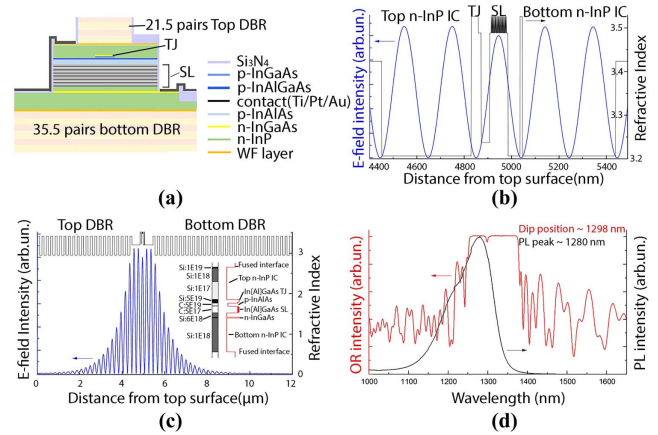


Fig. 8. (a) Schematic of the 1300 nm wafer-fused VCSEL; (b) field distribution and refractive index profile in the center; (c) field distribution and refractive index profile of the entire structure; (d) reflection spectra and PL spectrum of the device. Reprinted with permission from Ref. [103]. 2022, SPIE.

GaInAs QW layers to increase the photoluminescence (PL) wavelength, with the optional inclusion of antimony (Sb) to enhance growth morphology and optical efficiency by reducing group-III element surface diffusion^[105]. The English company IQE, in collaboration with the Technical University of Berlin, reported a 1300 nm dilute nitride VCSEL, which is fabricated on 76 mm GaAs wafers using molecular beam epitaxy (MBE), featuring a bottom DBR with 37.5 periods of silicon-doped AlGaAs, a 1 λ optical cavity with three GaInNAsSb QWs separated by GaAs barriers, and a top DBR with 18 periods of carbon-doped AlGaAs. The dilute nitride VCSEL was demonstrated with a bandwidth of 10 GHz and an error-free 12 Gbps bit rate across a 0.5 m standard OM1 multimode fiber^[106].

Strained InGaAs/GaAs QWs represent another efficient choice for the active region of GaAs-based VCSELs, demonstrating stable lasing and high modulation bandwidth at a wavelength as long as 1.1 μm ^[107]. Although room temperature lasing at 1.3 μm is demonstrated with highly strained InGaAs/GaAs double QWs^[108,109], the threshold current for these VCSELs is higher compared to conventional ones due to the large gain-cavity detuning, which limits their high-speed modulation performance. The reduced fiber attenuation at 1060 nm, which is less than half of that in the 850-nm band, combined with negative fiber dispersion, opens up the potential for high-speed data transmission over 10 km of standard SM fiber using 1060 nm strained InGaAs QWs VCSELs^[110]. We propose a 1060 nm VCSEL with intracavity surface relief engineering, as depicted in Fig. 9, which offers transverse resonance in coupled cavities and is designed to enable both single-mode operation and enhanced bandwidth. The VCSEL exhibited a single-mode power output of above 2.5 mW and a 3-dB modulation bandwidth of 23 GHz, along with successfully transmitting nonreturn to zero signals at rates up to 58 Gbps over a 5-km standard G652 SM fiber^[111]. Replacing QWs with quantum dots (QDs) in the active region of VCSELs offers another method for achieving long-wavelength GaAs-VCSELs, leveraging the tunability of

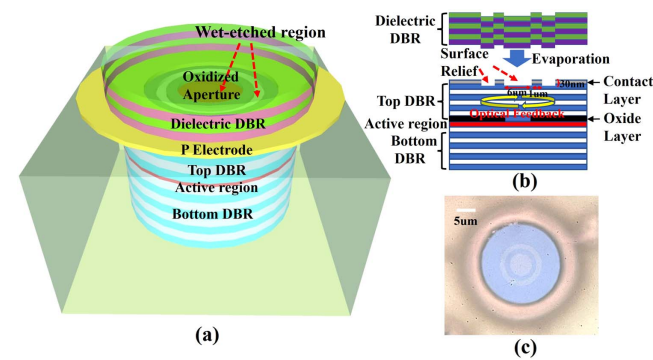


Fig. 9. (a) 3D Schematic of the 1060 nm surface relief VCSEL; (b) cross view of the device showing optical coupling; (c) top view of a fabricated device.

QDs’ emission wavelengths through adjustments in size and composition during fabrication^[112]. Utilizing MBE, room temperature lasing of VCSELs at 1300 nm has been successfully achieved with InGaAs/GaAs QDs^[113]. However, the low density of QD arrays necessitates the stacking of multiple rows of QDs to achieve the required optical gain, which restricts the modulation bandwidth of these QD VCSELs to a few gigahertz^[114].

Table 1 summarizes various representative works that have demonstrated high-speed, long-wavelength VCSELs for transmission back-to-back (BTB) or in fiber. As evident from the table, 1060 nm GaAs-based VCSELs are closest to commercialization for extended reach, offering stable lasing, ease of fabrication, and high-speed performance.

5. Advanced Packaging Technologies for VCSELs

While there have been significant recent improvements in the overall performance of VCSELs, the integration and packaging of optical modules are equally crucial in realizing practical interconnections. The quad small form-factor pluggable-double density (QSFP-DD) package mode stands as one of the

most popular choices in the design of high-speed optical transceivers^[120,121]. The QSFP-DD SR8 pluggable optical module featuring eight individual VCSELs, each capable of 50 Gbps PAM4 modulations, presents an effective solution for 400 G optical interconnects over distances up to 150 m^[122]. However, the traditional board edge pluggable module approach encounters a significant challenge in increasing link capacities due to the limited number of connectable optical modules^[123]. Efforts to shrink the footprint of a VCSEL-based optical engine^[124,125] to increase the number of connections per switchboard are majorly limited by power dissipation and internal physical area constraints. As link capacities in optical interconnects dramatically escalate in next-generation DCN and high-performance computing (HPC), this issue becomes even more severe, emphasizing the critical need for new high-density packaging solutions.

Co-packaged optics (CPO) technology, which integrates optical components closely with the main switching ASIC, offers significant benefits over traditional pluggable solutions, particularly by decreasing power consumption and increasing bandwidth density^[126–128]. Currently, the majority of CPO products rely on silicon-based optical solutions^[129,130], yet VCSEL technology presents a compelling option, noted for its cost-efficiency and reduced power consumption^[131]. The research group from the Chalmers University of Technology reported the successful integration of an 850 nm single transverse and polarization mode VCSEL with a silicon photonic integrated circuit (Si-PIC) employing SiN waveguides through tilted flip-chip integration over a grating coupler^[132]. Another notable development in this field is the collaboration between IBM and Finisar in the Project MOTION (multi-wavelength optical transceivers integrated on the node), which aims to create a VCSEL-based chip-scale optical module for direct attachment to an organic first-level package^[133]. In the MOTION optical package, the VCSEL chip is thermally linked to a copper shell and heat-spreader and assembled with the electrical and optical ICs on a 13 mm × 13 mm glass carrier. For the overall MOTION concept, a switch chip is surrounded by these 12

Table 1. Modulation Bandwidths and Bit Rates of VCSELs.

Group	Wavelength (nm)	Bandwidth (GHz)	Bit Rate (Gbps)	Distance (km)	Year	Ref.
Chalmers University of Technology	1060	22.5	40	2	2019	[115]
Georgia Institute of Technology	1060	20	100	0.1	2019	[116]
Tokyo Institute of Technology	1060	23	80	5	2022	[111]
IQE Corporate	1300	10	12	BTB	2020	[106]
Ioffe Institute	1300	8	25	5	2022	[103]
University of California at Berkeley	1550	7.8	10	100	2013	[117]
Connector Optics LLC	1550	9	30	BTB	2017	[118]
ITMO University	1550	13	37	BTB	2023	[119]

optical modules, each operating at 56 Gbps NRZ. In both the Tx and Rx of the electrical–optical–electrical (E-O-E) link, clock, and data recovery circuits are removed, as the signal integrity is sufficiently high to support a maximum distance of 30 m on OM4 fiber, which leads to a significant reduction in energy consumption. Researchers from the Hewlett Packard Enterprise (HPE) have developed a four-channel VCSEL-based CPO system utilizing CWDM technology, featuring VCSELs at wavelengths of 990, 1015, 1040, 1065, and 1090 nm^[134]. Figure 10(a) displays a schematic of the co-packaged CWDM optical module, where VCSEL and photodiode arrays are integrated and flip-chipped onto a high-speed organic substrate, achieving self-alignment during the solder reflow process. Lacking its own electrical chip design, HPE opts for an external detector featuring a 2D lensed InGaAs PIN photodiode array and assembles this with the optical module on a PCB equipped with a Huber Suhner MXP50 connector, as depicted in Fig. 10(b). Error-free transmission at 50 Gbps PAM4 per lane was achieved over a 100 m customized 1060 nm multimode fiber (MMF).

Bell Labs, in collaboration with the Eindhoven University of Technology, proposed a VCSEL-based CPO system which features a mode-multiplexed transmitter that supports 10 spatial modes by integrating a 2-D 10 G-class VCSEL matrix at C-band, arranged in a triangular layout, with a multi-plane light converter (MPLC)^[135]. This system employs 2.5-D integration technology to electrically connect BiCMOS driver ICs to a 2D array of single-mode C-band VCSELs via a patterned silicon interposer, as shown in Fig. 11. The MPLC, supporting 10 spatial modes, transforms VCSEL outputs for efficient transmission over fiber. They successfully demonstrated a 10-mode-multiplexed 200-Gbps line-rate transmission over 100 m of OM3 MMF and 28 km of graded index few-mode fiber (GI-FMF) using coherent detection.

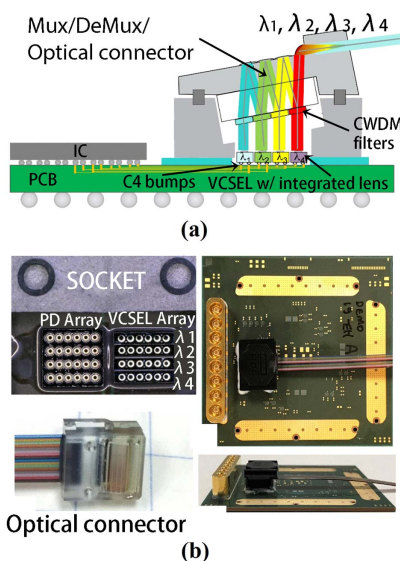


Fig. 10. (a) Schematic of the CPO CWDM optical module; (b) CPO module on a PCB. Reprinted with permission from Ref. [134]. 2020, IEEE.

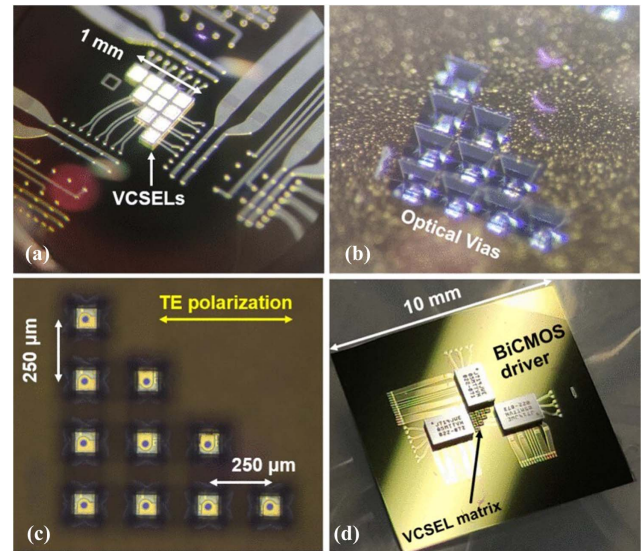


Fig. 11. (a) Top view of the flip-chip bonded VCSEL matrix; (b) bottom view of the VCSEL matrix; (c) micro-photo showing 250-μm spaced VCSEL apertures; (d) photo of the CPO module. Reprinted with permission from Ref. [135]. 2022, IEEE.

The research group from the Tokyo Institute of Technology has developed a high-capacity VCSEL-based CPO system^[136,137] utilizing a 16-channel bottom-emitting metal-aperture (MA) VCSEL array paired with a 19-core multicore fiber (MCF). The 16-channel bottom-emitting VCSEL array with an MA structure is shown in Fig. 12(a), achieving single-mode operations across all channels with 6 μm oxidation apertures.

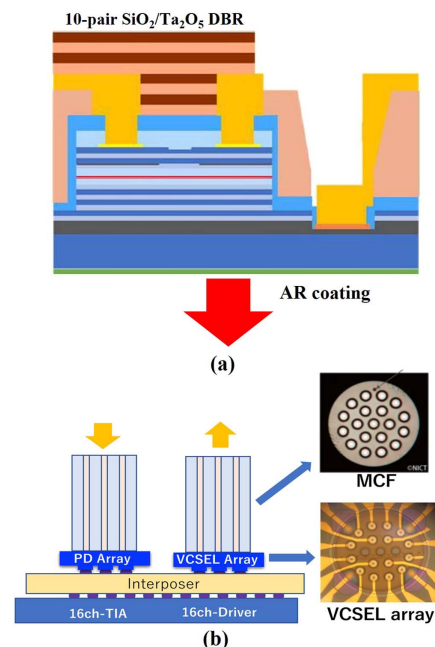


Fig. 12. (a) Schematic of the bottom-emitting VCSEL; (b) schematic of the VCSEL CPO transceiver. Reprinted with permission from Ref. [137]. 2023, OSA.

The system's total modulation speed surpasses 400 Gbps, attributed to the implementation of space-division multiplexing. After transmitting over a 5 km SMF, the system achieved transmission rates of 50 Gbps with NRZ modulation and 70 Gbps with PAM4 modulation per channel. The CPO transceiver packaging with 19-core MCF is carried out by Fujitsu Company^[138]. Figure 12 presents a cross-sectional view of a novel packaging structure for the optical transceiver module, designed for high-density optical interfacing. The structure includes a bottom-emitting VCSEL array and a back-illuminated PD array placed on an interposer's top surface, with an MCF connected through a butt-joint at the back of each optical component. Additionally, a 16-channel driver and TIA ICs are mounted on the interposer's back surface. This structure includes high-speed electrical wiring through the interposer's layers, leading to a narrow-pitch land grid array (LGA) as the module's electrical interface. The module housing is remarkably compact, measuring 7.8 mm × 16 mm × 8.0 mm, approximately the size of a coin.

6. Conclusion

In this comprehensive review, we have examined the latest advancements in high-speed, single-mode VCSELs, emphasizing their critical role in the evolution of next-generation DCNs. We delve into various aspects of VCSEL design and optimization, highlighting how these improvements are essential for enhancing modulation speeds and energy efficiency. This is particularly relevant, given the increasing data demands in DCNs. Additionally, our review underscores significant research efforts in single-mode control and the development of long-wavelength VCSELs, crucial for tackling the challenges of long-distance optical communication. We also explore the advancements in packaging technologies, especially CPO, showing their potential in revolutionizing optical interconnect solutions by offering high-density and energy-efficient options suited for the rapidly evolving needs of hyperscale data centers and contemporary DCNs.

Concluding with a historical perspective, the remarkable attributes of modern high-speed VCSELs—their compact size, high efficiency, and rapid operation, make VCSELs indispensable in the evolution of data communication technologies. In the transition to an era where data centers and global digital networks require greater agility and higher bandwidth, researchers worldwide are continuously advancing VCSEL technologies to address newly emerging challenges. As we look toward the future, these enhancements in VCSELs promise to drive future advancements in optical interconnects and will contribute significantly to the scalability, efficiency, and sustainability of hyperscale data centers and the global digital infrastructure.

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